LESSON PLAN

TEACHER : MR. NTALASHA…….. DATE: ……...

FORM : 1A…….. TIME: ……..…….

SUBJECT : MATHEMATICS…….. DURATION: ….……

TOPIC : 1.3 APPROXIMATIONS & ESTIMATIONS TOTAL ENROL = ……...

SUB-TOPIC : 1.3.2 SCIENTIFIC NOTATION…….. No. Of PPs: B= …….G=……

RESOURCES : STRINGS, TAPE MEASURE, RULER, BOARD RULER, MEASURING CYLINDER…….

REFERENCES : . O-Level Maths Syllabus Form 1, Pgs. 18 – 20, F1 Module Pg. 54 – 56 , CHAT GPT…

SPECIFIC COMPETENCES.

1.3.1.1 Use approximation in making informed decisions in real life.……

EXPECTED STANDARD:

Approximations used to make informed decisions appropriately.……...

PRE-REQUISITE KNOWLEDGE: Learners will have a basic understanding of whole and decimal……

numbers.……

RATIONALE : This lesson on "***Approximations and Estimations***" will equip pupils with critical skills. Used in everyday life to simplify complex measurements, ensure precision, and support informed decision-making. Teacher exposition, Group Work, Presentation methods will be used to present this lesson. This is a sixth lesson in the series of eight lessons.……

STAGES	TEACHER'S ACTIVITY	PUPIL'S ACTIVITY	METHODOLOGY	TIME
LESSON INTRODUCTION	- ***What numbers are represented by the following scientific notation.*** ***a)*** 3.123×10^3 ***b)*** 9.7×10^7 - ***Write down the following numbers in scientific notation.*** ***a)*** ***0.00000000657*** ***b)*** ***7560000000***	**Expected Response** **1) a) 3123** **b) 97000000** **2) a)** 6.57×10^{-9} **b)** 7.56×10^9	-Question & Answer	10 Mins
LESSON DEVELOPMENT	**Activity 6.1: Pair Work** 1) "A spacecraft traveling to Mars covers a distance of 225,623,000 kilometers. Engineers need to plan fuel requirements and communicate efficiently. a) Express this distance in scientific notation and round it to 2 and 3 significant figures?" b) If the cost of fuel per litre is K34.67 and the spacecraft consume 67000litres per km to reach Mars, determine the cost of the fuel required in standard form correct to 2 and 3 significant figures. c) State which cost is realistic for this distance and explain why.	- Be in pairs as guided by the teacher and answer the given questions. Expected Answers a) 2.3×10^8 (2 s.f.) 2.26×10^8 (3 s.f.) b) $1litre = K34.67$ $67000 = x$ $x = K2322890$ per km. $225623000 \times K2322890$ $C = K(5.2 \times 10^{14})$ (2 s.f.) $C = K(5.24 \times 10^{14})$ (3 s.f.) c) $K(5.24 \times 10^{10})$ is more realistic because it has more significant figures than the first one.	-Pair Work	15

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LESSON DEVELOPMENT 2	**Activity 6.2: Individual Activity** Answer the following: 1) The mass of the earth is approximated as being 5,972,000,000,000,000,000,000,000 kilograms. a. Express this in scientific Notation correct to i. 2 significant figures ii. 3significant figures iii. 4 significant figures b. If you were tasked with choosing one of the three notations in (a), which one would you choose and why? 2) Zambia's population is approximately 2×10^7 people. If each person uses 50 litres of water daily: a. Calculate the total daily water usage in litres, express your answer in **scientific notation.** b. If 1 litre is 1×10^{-3} cubic metres, convert the total daily water usage to cubic metres. Express the solution to a degree of accuracy of your choice. Explain why you have decided to use that degree of accuracy.	**Expected Solutions** **1) a) (i) 6.0×10^{24} (2 s.f.)** **(ii) 5.97×10^{24} (3 s.f.)** **(iii) 5.972×10^{24} (4 s.f.)** (b) I would choose (iii), because it shows more details than (i) and (ii). 2) a) 1000000000 **(1.0×10^9) *litres*** b) $(1 \times 10^{-3})(1 \times 10^9)$ $(1 \times 10^{9+(-3)}$ **(1.0×10^6) m^3** I have used 2 significant figures because it is much easier to interpret than with many significant figures.	Individual Activity	25
CONCLUSION	Ask learners to answer the following questions. 1) The concentration of a drug in a solution is 0.00000378 grams per milliliter. Write this concentration in scientific notation, rounded to **3 s.f.** 2) The mass of a dust particle in space is approximately 0.000000000000000000000000 00015 kilograms. Express this mass in standard form, correct to **1 decimal place**.	**Expected Responses** **1) 3.78×10^{-6}** **2) 1.5×10^{-27}**	-Individual Practice	

LESSON EVALUATION: ……..

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